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RESEARCH ARTICLE

People and Management Debt in ML-Integrated Software Projects: Structuring Industry Insights

PELIN DAYAN-AKMAN, ÖZDEN ÖZCAN-TOP, AND TUGBA TASKAYA TEMIZEL, (Member, IEEE)
Graduate School of Informatics, Middle East Technical University, 06800 Ankara, Türkiye

Corresponding author: Pelin Dayan-Akman (pelin.dayan@metu.edu.tr)

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ABSTRACT The accelerated development of Machine Learning (ML) tools, combined with broader access to frameworks and infrastructures, has driven the rapid adoption of ML-based solutions in industry. However, their integration into software systems introduces unique challenges, particularly for managing technical debt (TD). Traditional TD research focuses primarily on technical issues, but in ML systems, people and management factors, referred to as nontechnical debt (NTD), play a critical role in TD accumulation and persistence. In this study, we investigate the underexplored dimension of NTD in ML-integrated software systems, focusing on people- and management-related factors. Using Design Science Research (DSR) methodology, we developed an artifact in an iterative incremental manner that categorizes NTD issues in ML systems. As part of this process, we conducted semi-structured interviews with 18 professionals from 15 companies, examining 22 ML projects. Through thematic analysis, we identified 15 NTD categories, 10 of which relate to people debt, and the remaining 5 to management debt. Each category is associated with underlying causes, short-term fixes, and potential solutions. Our findings show that NTD in ML projects frequently arise from inadequate decision-making practices, particularly those related to technology adoption, knowledge management, and human resource planning. Additional sources of NTD include challenges in team dynamics, such as insufficient collaboration, poor skill integration, and ineffective team structuring, as well as communication barriers rooted in organizational culture and team interactions. These factors collectively and substantially impact project outcomes. While band-aid solutions may provide short-term relief, they frequently contribute to accumulation over time. To support practitioners and researchers, our study complements the proposed artifact with actionable recommendations informed by expert perspectives and literature.

INDEX TERMS Artificial intelligence, machine learning, machine learning life cycle, technical debt, nontechnical debt, people debt, management debt.

I. INTRODUCTION

Machine Learning (ML) technologies have become an integral part of software development, transforming how organizations solve problems through data-driven decision-making and automation [1], [2], [3]. While ML offers substantial opportunities to drive innovation and tackle complex problems, its unique characteristics, such as dependency on high-quality data, the need for continuous model updates, and reliance on evolving algorithms and frameworks, also introduce novel challenges.

In product development, achieving successful project results requires understanding where potential issues may arise, identifying their sources, and analyzing their

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underlying causes. Although several life cycle models, such as CRISP-DM [4] and the Team Data Science Process (TDSP) [5], and standards such as ISO/IEC 5338 Artificial Intelligence System Life Cycle Processes [6], provide guidance for ML development, they fail to systematically identify and categorize recurring development issues in ML-integrated systems.

In AI/ML development, unresolved, recurring issues cause technical debt (TD) [7], a concept that is widely used to describe the long-term consequences of short-term decisions in software engineering [8]. Recognizing this, we adopted a TD-oriented perspective to examine ML-specific issues and structure them in a way that supports proactive identification, analysis, and management of TD.

Technical debt (TD) is defined as the accumulation of suboptimal design or implementation decisions made during software development [8], [9], [10]. Although such decisions may expedite development in the short term, they often lead to increased costs and effort for future maintenance, scalability, and extension. TD has been extensively studied in traditional software development, particularly from a technical perspective [11], [12], [13], [14]. These studies focused on concerns such as architectural complexity [15], ensuring maintainable and reusable code [14], and developing robust testing strategies [16], which are relevant for both traditional and ML-integrated software development.

In addition to technical elements, factors related to individuals, processes, culture, and social dynamics, called nontechnical debt (NTD) [17], also play a significant role in the accumulation and persistence of TD. These aspects highlight the interplay between technical and nontechnical influences on software quality. Furthermore, ML systems introduce unique complexities to TD management because of their distinct characteristics such as data dependencies, the experimental nature of model development, and the evolving requirements of integrating ML into production environments.

The existing literature on TD in ML systems is relatively limited and focuses primarily on technical dimensions. In ML contexts, specific issues include TD arising from data dependencies, such as inconsistencies or shifts in data distributions, and the need for regular retraining and updating of models to maintain performance over time [7], [18], [19]. Moreover, the modularity of ML pipelines, the integration of evolving algorithms, and the experimental nature of model development further complicate the TD landscape in ML projects [20], [21], [22]. These studies provide valuable insights into the mitigation of technical aspects of debt, helping to improve system reliability and scalability. However, an equally important yet underexplored dimension is the role of human factors in the emergence and management of TD in ML-integrated software systems. These factors, such as decision-making processes, team dynamics, and communication challenges, often serve as root causes that influence how TD accumulates and evolves in ML systems.

Human aspects such as development team decision-making processes, communication gaps, and the dynamics of cross-disciplinary collaboration in ML projects often play a crucial role in the development and accumulation of TD. These factors uniquely contribute to TD by influencing design decisions, prioritizing tasks, and managing project constraints. Despite their significance, these aspects are often overlooked in favor of technical challenges. Addressing these human factors is essential for gaining a comprehensive understanding of TD [23], particularly in ML systems, where collaboration among software engineers, data scientists, and domain experts is critical [24].

In this study, we focus on the TD that arises from people and management aspects of ML-integrated software systems. Our aim is to provide a structured categorization of NTD for ML-based projects, examining how people and management-related elements contribute to NTD accumulation and persistence across various stages/processes of the ML project life cycle. This includes identifying NTD cases, their root causes, as well as potential band-aid and suggested solutions.

To achieve this purpose, we adopted the Design Science Research (DSR) methodology and conducted semi-structured interviews with 18 industry professionals from 15 companies, encompassing 22 projects. The collected data were analyzed using thematic analysis to identify TD-related issues. Through iterative refinement and expert reviews, we developed a robust categorization of people- and management-specific TD issues in ML-integrated software systems.

This research is part of a broader study, including a total of 53 categories identified, that examines both TD and NTD cases throughout the entire ML life cycle, including data preparation, model development, deployment, and maintenance stages. However, this research specifically addresses **the people- and management-related dimensions of TD** in ML-integrated software systems.

Our key contributions are as follows:

- We systematically analyzed **15 NTD categories related to human, organizational, and strategic factors**, shedding light on the underexplored dimension of NTD in ML projects.
- We identified **the underlying causes** of these NTD cases and examined **commonly applied band-aid solutions**, illustrating how short-term fixes often result in new or compounding forms of NTD.
- We presented a set of **practical solution strategies** for NTD cases derived both from expert interviews and recent academic literature.
- To our knowledge this is the first study to systematically structure NTD related to people and management in ML-integrated software systems based on interview data collected from projects across a diverse range of topics.

The remainder of this paper is organized as follows. Section II reviews the related work, and Section III outlines the research design. Section IV provides the demographics of

the participants, projects, and companies. Section V presents the artifact, followed by a discussion in Section VI, which places our findings in the context of the existing literature and expert reviews. Section VII addresses threats to validity, and Section VIII concludes the paper.

II. BACKGROUND AND RELATED WORK

This section outlines prior research on TD. The three subsections address TD in conventional and ML-integrated software development and NTD in ML-integrated software projects, respectively. The final subsection also presents existing work on NTD together with research gaps that inform the contribution of this study.

1) TECHNICAL DEBT IN CONVENTIONAL SOFTWARE DEVELOPMENT

TD describes how short-term coding shortcuts can lead to long-term maintenance challenges and increased costs [8], [25]. Over time, TD software has been categorized into different types, such as code debt, design debt, and test debt [16]. However; it is argued that most research studies focus mainly on code-related debt, often overlooking other types of debt such as architectural, design, and documentation debt, as well as TD management from broader perspectives [26]. Focusing predominantly on code debt may likely originate from the initial definition of TD, which is centered on code-level issues. However, research over time has shown a significant impact of human and organizational factors on technical decisions, stating that these factors are as crucial as technical expertise. Kruchten et al. [27] studied the causes and implications of TD, and examined its progression from a simple metaphor to a well-defined and actionable framework in software engineering. Their research emphasized the importance of moving beyond viewing TD as merely a collection of coding shortcuts. Instead, they proposed understanding it as a complex and multifaceted concept, influenced by a range of technical and organizational factors. They provided a thorough analysis of the dimensions of TD and traced its origins to pressures for rapid software delivery and trade-offs often made during the software development life cycle. Kruchten et al. [27] also proposed a framework for managing TD, emphasizing the need for regular code refactoring and comprehensive design reviews to mitigate its long-term impact.

2) TECHNICAL DEBT IN ML-INTEGRATED SOFTWARE PROJECTS

The concept of TD in ML projects presents additional complexities due to the distinctive aspects of the ML life cycle, such as extensive data dependencies and the challenges of deploying models into production. Sculley et al. [7] were pioneers in addressing these challenges, highlighting that TD in ML systems often remains hidden. They identified specific TD, including pipeline jungles, glue code, and configuration debt, that pose significant obstacles to the scalability and maintainability of ML systems. Subsequent research examined various dimensions of TD in ML systems, focusing on challenges and potential solutions at different stages of the ML life cycle [19], [28], [29]. These studies span several key areas, including issues unique to Deep Learning (DL) applications, the pivotal role of data quality and management, and the shift to structured operational practices such as MLOps [29] and DataOps [30].

3) NONTECHNICAL DEBT IN ML-INTEGRATED SOFTWARE PROJECTS

While human and organizational aspects are often classified as TD, Saeeda et al. [17] emphasized the importance of recognizing NTD as a separate concept, drawing attention to its broader impact on organizational and social dynamics. Drawing from 40 sources in both the white and gray literature, this study identifies five types of NTD: Process Debt, People Debt, Culture Debt, Social Debt, and Organizational Debt. Process debt refers to inefficiencies such as bottlenecks or wasteful stages within a process. We consider process debt a foundational category that intersects both technical and nontechnical dimensions, potentially encompassing aspects of various TD types. People debt involves challenges that delay or hinder specific development operations and are categorized into behavioral challenges, work challenges, and issues related to collaboration, communication, and coordination. Culture debt arises from misaligned hires, weak organizational connections, unparticipatory practices, and managers' lack of cultural understanding. Social debt is linked to poor social structure, team pressure, inadequate communication, and global distance. These categories often overlap with no clear boundaries and apply to all software systems, including those integrated with ML.

Although human aspects have been explored in software engineering, research focusing on these elements in ML-based systems is still scarce. Researchers have pointed out that skill gaps are a significant factor contributing to TD in ML projects. Challenges such as the need for deep domain expertise and advanced knowledge of DL models often exceed the capabilities of existing teams [30], [31]. Furthermore, the limited availability and high cost of skilled professionals, such as data scientists, data engineers, and subject matter experts, create significant challenges for organizations. Competition for these experts is intense, and major technology companies often secure top talent, making it difficult for others to build and retain capable teams [32].

Challenges also appear in the ethical and governance aspects of AI. The lack of uniform standards for accountability can lead to complicated TD situations that may obstruct the adoption of responsible AI practices and interfere with meeting regulatory requirements [33], [34]. For example, although tools and frameworks designed to enhance transparency and mitigate risks in AI systems provide valuable guidance, they often do not fully address social biases or potential harm. Addressing these challenges effectively requires collaboration between diverse fields,

as the underlying issues are shaped by complex social, historical, and contextual factors that extend beyond purely technical considerations [35]. This underscores the importance of interdisciplinary approaches to fostering ethical and responsible AI development.

Issues such as lack of management support, team misalignment, and resource constraints are also well documented in traditional software development, becoming more evident in ML contexts [17], [24]. Inadequate teamwork, communication gaps, and poor collaboration often lead to misaligned processes and repeated efforts [17]. Misaligned team structures, particularly in multi-organization setups, can amplify challenges such as a lack of standardization and coordination, complicating model deployment and maintenance [29]. Communication bottlenecks between data scientists and operational teams also contribute to inefficiencies, delays in timelines, and compounding of TD. Although significant research has been conducted on TD in conventional software systems, TD in ML-specific contexts remains underexplored, particularly in relation to people, organizational, and management factors. The understanding of TD has expanded to include sociotechnical dimensions by acknowledging the role of human factors along with technical aspects. This perspective emphasizes that decisions, practices, and team dynamics significantly influence the accumulation of TD. Addressing NTD is now considered an essential part of comprehensive TD management in ML-based software systems.

III. RESEARCH DESIGN

Figure 1 illustrates the Design Science Research (DSR) methodology [36] we adopted, which aligns effectively with our research goal of developing a structured artifact for categorizing TD from people and management-related perspectives in ML-based software development projects. DSR was selected due to its focus on the creation and iterative refinement of practical artifacts to address identified problems in real-world contexts [37]. Although elements of Grounded Theory, specifically thematic analysis and coding techniques, were incorporated to analyze qualitative data [38], the overall objective of systematically developing and validating an artifact guided our selection of DSR over theory-generating methodologies.

Our research process follows the model proposed by Peffers et al. [36], which offers a step-by-step structure for artifact development, and is aligned with the foundational principles of Hevner et al.'s Information Systems Research Framework [39]. As illustrated in Figure 1, itProblem Identification and *Definition of Design Objectives* phases address the **relevance** dimension in Hevner's framework by grounding the study in real-world organizational and technical challenges. The *Design and Development* phase reflects the **design cycle**, during which the artifact was iteratively constructed and refined. The *Evaluation* phase corresponds to the **rigor** dimension of Hevner's framework, supported by multiple rounds of expert and assistant reviews.

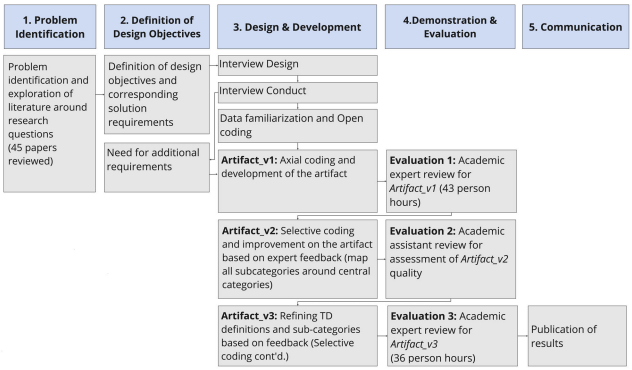

FIGURE 1. Design science research process.

Finally, the *Communication* phase ensures contribution to the **knowledge base** through the dissemination of the results.

A. PROBLEM IDENTIFICATION

The initial phase involved identifying and defining the research problem. Despite existing life cycle models (e.g., CRISP-DM and TDSP) and standards (e.g. ISO/IEC 5338 AI System Life Cycle Processes), significant gaps persist regarding the proactive identification and management of people and management-related TD in ML-integrated software systems. Through a comprehensive literature review, we formulated the following research questions (RQs) to address these gaps:

RQ1: What are the characteristics and definitions of TD related to people and management factors in ML-specific projects and what impact do these factors have?

RQ2: What people and management-related causes lead to the incurrence of TD in ML-specific development projects?

RQ3: What people and management-related band-aid solutions contribute to the recurrence of TD in ML-specific projects?

B. DEFINITION OF DESIGN OBJECTIVES

Based on these research questions, we derived two specific design objectives (DOs) to guide the development and evaluation of our artifact.

DO1: Comprehensive Categorization of People and Management-related TD (Aligned with RQ1, RQ2, and RQ3): The artifact should offer a systematic and detailed categorization of people and management-related TD in ML projects, clearly capturing and defining the characteristics, underlying causes, and common band-aid solutions contributing to recurrent TD.

DO2: Practical Guidance for TD Mitigation (Aligned with RQ4): The artifact must be actionable, providing practitioners with effective, evidence-based strategies to proactively manage, reduce, or eliminate people and management-related TD. These solutions must be practical, grounded in real-world insights from experts, and validated through rigorous evaluation.

C. DESIGN AND DEVELOPMENT

As presented in Figure 1, this phase focused on the design and development of the artifact including the categorization of TD and a structure that allows systematic data categorization. The final artifact was developed in an iterative incremental manner. Data used in this study were collected through semi-structured interviews. The interviews targeted participants with various roles related to ML and varying levels of management to ensure comprehensive and representative insights.

In the following subsections, we provide detailed explanations of the interview process, including participant selection, sampling strategy, interview design, interview conduct, and additional stages, including thematic analysis and artifact development.

1) INTERVIEW DESIGN

The interview design aimed to comprehensively explore the multifaceted concept of TD within ML-integrated software projects. Considering the complexities involved in understanding TD, we selected semi-structured interviews to allow flexibility and depth in exploring the participants' experiences [40]. We designed the interview questions based on the findings of the literature review, structured around key phases of the ML development life cycle to ensure systematic exploration of TD issues.

Specifically, the interview flow addressed the following stages:

- *Data Preparation*
- *Model Development*
- *Model Deployment*
- *Software and Hardware Tool Management, and Configuration Management*
- *People, Culture and Strategy*
- *Strategic Alignment*
- *Organizational Collaboration*

Given that TD often compounds over time, we planned to discuss projects that had at least reached the deployment phase and were completed within the last three years. This criterion aimed to ensure that the discussions were relevant and that the issues described were recent enough to provide valuable insights into current practices in ML projects.

2) PARTICIPANT SELECTION AND SAMPLING STRATEGY

The participants were strategically selected to represent diverse ML roles, such as data scientists and ML engineers, across various organizational domains and levels. Special emphasis was placed on including high-level decision-makers to capture critical managerial insights. Additionally, underrepresented domains were specifically targeted to enhance representativeness and mitigate sampling bias [41]. Furthermore, participants who worked with varying types of data were deliberately included to enrich the diversity of their perspectives.

Strategic participant selection expanded beyond immediate availability, involving coordinated scheduling and the use of professional networks to reach second-degree contacts, thus enriching the diversity and comprehensiveness of the participant pool to capture multifaceted perspectives on TD. To ensure thematic saturation, the interviews continued until no new TD categories emerged.

3) INTERVIEW CONDUCT

Before interviews, participants received personalized invitations and informed consent forms detailing the purpose of the research, the expected duration of the interview, and the rights of participants. Each interview began with the collection of demographic information (Table 3), contextualizing the insights of the participants according to their organizational roles and the specifics of the ML project they would be discussing.

To accommodate participants less familiar with technical jargon, we avoided using the term "technical debt", focusing instead on general "issues", their root causes, and resulting suboptimal decisions.

Memoing, defined as the ongoing documentation of insights and emerging ideas during data collection [42], were extensively used during the interviews. Specifically, if participants did not spontaneously provide root causes or related solutions, follow-up probing questions were posed based on these memos to ensure completeness.

The first interview was conducted as a pilot study to assess the length and clarity of the questions. It also provided an opportunity to follow up with the participant for additional clarification. Based on this interview, we identified areas in which the questions could be improved. For example, we revised the complex items by splitting them into simpler parts without altering their context, which led to more focused and detailed responses by enhancing participant understanding.

The revised interview structure first asked the participants to describe the problems they encountered and their root causes during each phase of their ML projects. After that, we inquired about the suboptimal solutions they had applied and thoroughly explored each topic before proceeding. The interviews followed a structured design aligned with the stages of the ML life cycle, as outlined in Section III-C1.

The interviews were conducted in the participants' native languages (English or Turkish) and recorded with their consent, ensuring accurate transcription for subsequent detailed thematic analysis.

4) FAMILIARIZATION WITH THE DATA AND OPEN CODING

After the interviews, the recorded sessions were transcribed using Google Speech-to-Text,[1] a speech recognition model to automate the transcription process, which produced initial drafts with features such as dual-language support (English-Turkish) and automatic punctuation to enhance accuracy. The

[1] https://cloud.google.com/speech-to-text

transcripts were then reviewed and the unclear parts were clarified using the original audio. We analyzed the transcripts in Dedoose application,[2] a qualitative research tool designed to code and identify patterns and themes.

The coding process in thematic analysis involved systematically labeling text segments with codes that represented key pieces of information relevant to TD in ML projects. Following an open coding approach [43], the transcripts were broken down into small, meaningful segments, such as phrases, and each segment was assigned a descriptive conceptual label. This approach enabled the identification of conceptually similar events, which were then grouped into preliminary categories and subcategories. Therefore, nine specific types of information were constructed:

1) The identified issue subject to TD.
2) The root cause of the issue.
3) The impact or consequences of the issue.
4) The suboptimal solutions causing recursive TD.
5) The root cause of the suboptimal solution.
6) The impact of the suboptimal solution.
7) The suggested solutions eventually implemented.
8) The reasoning behind choosing the suggested solution.
9) Solutions that were successful but unrelated to TD.

The ML life cycle stages and the elements identified during the interview design phase were used as the primary structure for this coding process. The main parent codes were the TD issues, suboptimal solutions, and suggested solutions, each encompassing several child codes corresponding to different ML life cycle stages. Each case was assigned a unique code and its associated root causes were coded as child codes in the relevant case.

Data familiarization was ensured at the participant level, meaning that if multiple participants mentioned similar problems, their insights were included under the existing case ID while noting the specific participant ID. This approach helped maintain a cohesive data set and prevented redundancy.

5) AXIAL CODING AND ARTIFACT DEVELOPMENT: GENERATING INITIAL THEMES AND CATEGORIZATION STRUCTURE (ARTIFACT_v1)

The initial thematic structure was generated by clustering the coded segments into higher-level categories that reflect the recurring patterns in TD related to people and management. This stage involved axial coding [43], where connections between codes were identified and grouped into broader categories to reveal underlying structures in the data.

Our approach simultaneously developed both the categorization of TD and a structure that allows systematic data categorization. This dual outcome justified the use of the DSR methodology and was designated Artifact_v1.

This iterative coding and thematic structuring process underwent an iterative refinement informed by expert evaluations, resulting in Artifact_v2 and Artifact_v3. These iterations progressively enhanced thematic coherence, contextual accuracy, and practical applicability, thus confirming artifact validity and usability, as explained in the following sections.

6) SELECTIVE CODING AND ARTIFACT DEVELOPMENT: REVIEWING AND DEVELOPING THEMES AND CATEGORIZATION STRUCTURE (ARTIFACT_v2)

An iterative process refined the initial themes by examining their coherence and relevance, adjusting the categories based on feedback, and identifying emerging patterns through memoing and expert input. The development and evaluation of the artifact occurred through several iterative and incremental cycles, involving academic experts from the fields of ML and Software Engineering integral to both processes. Both experts are also co-authors of this paper. During the evaluation process, academic experts provided feedback that also informed updates during the development phase. One academic expert has been involved in Data Science (DS) and AI since 1999, providing consultancy in academia as well as to national and international organizations and companies. The other expert had over 15 years of expertise in software engineering and management, further enriching the development and evaluation phases.

In the first round of evaluation, Artifact_v1 was subjected to a thorough review, in which the experts provided detailed feedback on the categorization of TD and the organization of the cases. At the conclusion of the development and evaluation of the artifact within the first iteration, academic experts dedicated more than 43 person-hours to this phase. Feedback gathered during this review highlighted areas for improvement, particularly in refining the categorization structure. This new structure ensured that root causes and contextual details were adequately captured.

The categorization was refined to include categories that encapsulated the case within the most relevant context, whereas these categories were linked to broader, more generic categories. Band-aid solutions and suggested solutions were analyzed in the same context as the TD cases to preserve the narrative flow and ensure that the solutions were evaluated within the specific circumstances that gave rise to them. Because band-aid solutions often led to further TD, they were categorized using the same structure as the original TD cases. This stage reflects selective coding [43], where the aim was to refine and relate all subcategories around central categories, establishing coherence throughout the emerging structure. The discussion solutions of the researchers were revised accordingly, completing the development of Artifact_v2.

7) SELECTIVE CODING AND ARTIFACT DEVELOPMENT: REFINEMENT OF CATEGORIES (ARTIFACT_v3)

In the final stage, the distribution of categories across the entire artifact was summarized. Overly broad or nonspecific categories that did not refer to specific TD were categorized. Conversely, categories that were too specific and redundant were consolidated into more general categories. This step

[2]https://www.dedoose.com/

continued the selective coding process [43], focusing on refinement of category boundaries. This refinement process ended with the development of Artifact_v3. Artifact_v3 was then subjected to a second round of expert review. In this round, the entire set of cases is reassessed. The experts meticulously evaluated the alignment of the categorization with the cases, clarity, and precision of the definitions.

The iterative review process ended with the finalization of the artifact, which incorporated the insights and refinements suggested by academic experts. At the conclusion of the development and evaluation of the final artifact in the second round, academic experts dedicated 36 person-hours to this phase. Successful completion of this evaluation phase confirmed the validity of the artifact. The first round of expert evaluation focused on refining artifact clarity, comprehensiveness, and contextualization of categories, whereas the second round emphasized validating the refined structure, precision, and stronger linkage between categories and cases.

The iterative process of thematic analysis and artifact development, as depicted in the DSR process diagram in Figure 1, highlights the evolution of categorization and structure from its initial form to a refined and comprehensive artifact capable of capturing the complexity and nuances of TD in ML projects. This study represents a dynamic artifact that emerged during the research process. This artifact, including the TD categorization, was iteratively developed and refined based on the data collected and the expert feedback received throughout the study. Consequently, it serves as the final artifact produced by this research and provides a basis for future exploration and development.

D. DEMONSTRATION AND ARTIFACT EVALUATION

An assessment and scoring of data quality was performed for artifact refinement. The artifact, the ML TD categories, and the definitions presented in Section V, were systematically validated by an independent researcher from the study team who had no involvement in its development. Using a 5-point Likert scale, the validation assessed the dataset on the criteria of understandability and level of detail, as described in the work of Sonnenberg and Vom Brocke [44]. The results showed that only 3% of the cases scored less than five for the understandability criterion, while 5% fell below five for the Level of Detail criterion.

IV. DEMOGRAPHICS OF PARTICIPANTS

The interviews were conducted with 18 participants from companies of varying sizes, ranging from small firms with 1-50 employees to large enterprises with over 10,000 employees located in different countries such as Türkiye, Germany, the US, the Netherlands, and Bulgaria, as shown in Table 1. This demonstrates that the data collection involved diverse and significant industry players.

The interviewees represented a diverse range of roles and experiences in ML and DS, as shown in Table 3. The interviewees held various senior positions, including Vice Presidents, Directors, and Department Heads, as well as specialized roles, such as Senior Computer Vision Engineers and Computational Linguists. This range highlighted the inclusion of professionals with both managerial and hands-on technical expertise.

TABLE 1. Company demographic details.

Company ID	Company Size	Country
C1	201-500	Türkiye
C2	51-200	Germany
C3	10,000+	Netherlands
C4	51-200	Türkiye
C5	1-50	Bulgaria
C6	1-50	Türkiye
C7	501-1,000	Türkiye
C8	10,000+	Türkiye
C9	51-200	Türkiye
C10	1,001-5,000	Türkiye
C11	201-500	US
C12	51-200	Germany
C13	10,000+	US
C14	1,001-5,000	Türkiye
C15	51-200	US

A total of 22 projects were investigated, as shown in Table 2, covering a wide range of business domains such as Security, Finance, Retail, Automotive, Educational Technology, and Public Safety, showcasing a diverse set of ML applications, from image recognition to content recommendation.

In terms of educational background, the majority of the participants held advanced degrees (Master's or Ph.D.) in fields closely related to ML and DS, such as computer engineering and information systems. This high level of education aligns with their considerable experience in ML and DS projects, with several participants completing more than 20 projects throughout their careers. The years of experience of the participants in the field also varied, from junior professionals with 1.5 years of experience to senior experts with more than 17 years of industry experience.

The interviews averaged 1 hour and 16 minutes, indicating in-depth discussions with each participant. The number of ML/DS projects currently being managed or maintained also varied, with some participants overseeing up to 8-10 projects at the time of the interviews. This diversity of experience provides a broad perspective on the challenges and solutions faced in ML projects across different organizations and domains.

V. THE ARTIFACT: PEOPLE AND MANAGEMENT DEBT IN AI-INTEGRATED SOFTWARE PROJECTS

This section presents the final version of the artifact (V3) that emerged from the iterative DSR process, reflecting the consolidated outputs and contributions developed across all DSR phases. Specifically, it introduces 10 categories of People Debt and 5 of Management Debt, detailed in Tables 4 and 5 and further discussed below. Each category is illustrated with case-specific findings, root causes, and, where available, band-aid and suggested solutions as shared

TABLE 2. Project demographic details (PJID = Project ID, CID = Company ID).

PJID	CID	Business Domain	ML Application Area	Aim and Scope
PJ1	C1	Security	Online Fraud Detection	Preventing spammer activity and protecting domain reputation by analyzing all daily emails.
PJ2	C1	Security	Online Fraud Detection	Detecting whether emails sent to customers were spam or legitimate, covering all daily outgoing emails.
PJ3	C1	Security	NLP-based Malware Filtering	Filtering malicious forms created on the company's product by detecting fraudulent or abusive uses, such as phishing and credential theft.
PJ4	C1	SaaS	Anomaly Detection on Time Series Data	Automating processes and improving efficiency by reducing stakeholder intervention through anomaly detection on time series data, focusing on other project teams within the company.
PJ5	C1	Security	Image Recognition	Detecting if images in outgoing emails sent via the company's product matched recognizable brands for potential fraud.
PJ6.1	C2	Mental Health	Content Recommendation	Helping users easily find the desired content, covering all company users.
PJ6.2	C2	Mental Health	Customer Conversion Prediction	Predicting purchases based on user behavior instead of LTV, accounting for post-COVID changes, focusing on company users in the first 7 days of the trial who had not made a purchase.
PJ7	C3	Customer Relationship Management	Time Series Forecasting	Forecasting annual customer service contact volumes to enable weekly scheduling, planning, and annual budgeting, covering all users and stakeholders, including planners, finance teams, and C-level executives.
PJ8	C4	Finance	Retention Prediction	Identifying customers likely to retain, focusing on the customer call center.
PJ9	C5	Sports Analytics	Data Analytics with Multiple Data Types	Providing sports managers and decision-makers with insights to make informed decisions on player transfers by estimating transfer values and evaluating return on investment.
PJ10.1	C6	Automotive	Image Recognition	Developing Advanced Driver Assistance Systems (ADAS) features, including lane keeping or collision warnings. The scope covered vehicles equipped with ADAS Level 2-3 systems.
PJ10.2	C7	Safety Critical	Image Recognition	Detecting drones from camera footage, primarily to prevent unauthorized drone activity in areas like airports.
PJ10.3	C8	Supply Chain Management	Demand Forecasting	Forecasting production quantities for parts manufactured for the company's subcontracted firms, incorporating COVID-era impacts and addressing changing manufacturer projections.
PJ11	C9	Safety Critical	Image Recognition	Detecting, classifying, and tracking surface objects, estimating their distance, and position, for integration into marine vessels across various sectors.
PJ12	C10	Public Safety	Image Recognition	Monitoring rule violations by bus drivers and measuring passenger density between stops using multiple DL models.
PJ13	C11	Educational Technology	Content Recommendation	Improving search functionality by using LLM-based semantic search to match user queries with course catalogs and ranking results effectively.
PJ14	C12	Retail Security	Image Recognition	Analyzing retail self-checkout transactions to detect errors or theft.
PJ15	C13	NLP	NLP Tool Development	Developing a syntactic parser for Turkish. The scope included anyone working on Turkish-based products or research.
PJ16	C14	Automotive	Image Recognition	Modeling the perception layer with AI for ADAS and autonomous driving to detect dynamic objects and behaviors, as well as stationary elements like road structures and traffic signs, for all planned vehicle productions.
PJ17	C15	Retail	Inference Optimization	Optimizing ML workloads on NVIDIA GPUs by converting deep neural network models to TensorRT. The scope included optimizing multiple interconnected DNN modules across various product pipelines to ensure efficient GPU execution.
PJ18.1	C14	Automotive	Image Recognition	Developing perception algorithms for detecting static objects, such as lanes and traffic signs. The scope included implementation in all planned vehicle productions.
PJ18.2	C10	Security	Image Recognition	Implementing a DL-based fingerprint matching algorithm for the customer, focusing on matching latent (unclear) fingerprints.

by participants. To present these findings clearly and effectively, we selectively included only the most striking and relevant participant quotations, ensuring that the examples are illustrative.

A. PEOPLE DEBT CATEGORIES AND ASSOCIATED CASES

In this section, we present 10 categories identified under People Debt. Each category highlights distinct aspects of people-related challenges that contribute to TD within projects.

1) TECHNOLOGY AND KNOWLEDGE ADAPTATION DEBT

This debt is characterized by challenges associated with resistance or insufficient adaptation to new technologies and knowledge within organizations, alongside reliance on outdated systems. The **root cause** was noted as the reluctance to adopt new ML technologies, particularly from technical teams other than the core ML project team, such as the DevOps team, as stated by participants P1 and P2. This case was observed in security domain projects (PJ1, PJ2), where security concerns are prioritized, as stated by P1, as follows:

> *"The backend doesn't allow much flexibility for library changes. This means that we cannot use the languages we prefer for our projects because it could create security vulnerabilities."* – P1 (VP of Data)

P9, P10, and P12 also reported that the disruption of existing workflows as a result of the introduction of new technologies and the desire to use familiar tools were other root causes of this case. As a **band-aid solution**, the ML team went ahead with testing and running their microservices for the developed ML tasks without approval of the DevOps team. They were aware that this might cause potential security vulnerabilities, as stated by P1. However, in such cases, when there is conflict among ML teams, the **suggested solution** would be to promote collaboration and communication within the technical teams to establish secure systems, as stated by P2.

Another notable case involved the difficulty of keeping up with rapid developments in the ML domain, as articulated by P5, where the **root cause** was the rapid evolution of ML technologies and the frequent introduction of superior models and tools, which often evolve faster than teams can adapt. This case was identified in Project PJ5, which focused on an image recognition application.

TABLE 3. Interview participants' demographic details (PID = Participant ID, LDG = last graduation department and degree, NC = Number of completed ML/DS projects, EF = Experience in ML/DS field (in years), ND = Number of ongoing ML/DS projects including maintenance).

PID	Current Role	LDG	NC	EF	ND
P1	VP of Data	Scientific Computing (MSc)	9	7	3
P2	Director of Data Science	Computer Engineering (BSc)	20	8	8-10
P3	Data Scientist	Big Data Analytics (MSc)	8-9	7	3
P4	Anomaly Detection Team Lead / Data Scientist	Computer Engineering (PhD)	11	5	6
P5	Jr. Data Scientist	Computer Engineering (BSc)	4	1.5	3
P6	Data Scientist	Mathematics (BSc)	4	6	-
P7	Data Scientist	Information Systems (PhD)	40	8	4
P8	Data Scientist	Industrial Engineering (BSc)	3	1.5	3
P9	Co-founder / CTO	Information Systems (PhD)	25-30	11	5
P10	Sr. Computer Vision Engineer	Computer Engineering (MSc)	7-8	8	1
P11	Head of Computer Vision, ML and Robotics	Information Systems (PhD)	10-15	17	8
P12	Machine Learning Engineer	Information Systems (MSc)	15-20	6	4
P13	Sr. Manager - Search Engineering	Electrical and Electronics Engineering (PhD)	43	10+	5
P14	Sr. Deep Learning Engineer	Information Systems (PhD)	5	12	4
P15	Computational Linguist	Cognitive Science (PhD)	15	16	-
P16	Machine Learning Engineer	Electrical and Electronics Engineering (MSc)	3-4	6	1
P17	Optimization Team Lead / Sr. Data Scientist	Computer Engineering (PhD)	-	5	3
P18	Machine Learning Engineer	Computer Science (MSc)	-	-	-

P13 emphasized the issue of rapidly adopting new ML technologies without considering available resources. This issue is **rooted in** upper managers' concerns about staying competitive in the market. It is further **driven by** a fear-of-missing-out (FOMO), which creates an urge to constantly experiment with new technologies.

The lack of an independent R&D team to develop and evaluate new technologies was another **root cause**.

> *"There's a tendency to adopt new technologies based on market trends. As a result, they often avoid investing in short- to mid-term research and instead aim to quickly adopt and use the technology."* – P13 (Senior Manager)

Inflating the successes of new technologies also misleads ML teams, adding to the challenge as a **root cause**, as stated by P13 as follows:

> *"We consider a technology to adopt because there is a lot of hype around it. In this case, while their experiments may show success, it might not be relevant to your specific domain."* – P13 (Senior Manager)

As a **band-aid solution**, proactive solutions were often delayed until problems appeared in newly adopted, untested ML solutions or libraries that were unknown to the ML team, as noted by P13.

> *"The team adopted a new solution without fully understanding its conditions. After it failed in production, we spent a lot of time cleaning it. The reactive approach delayed the detection of the issue and made it worse."* – P13 (Senior Manager)

Another **band-aid solution** involved attempts to standardize the technology stack across teams. However, this led to requirement conflicts, as different teams were assigned incompatible mandates by upper management.

In such a case, the **suggested solution** would be to establish an independent R&D team to evaluate new technologies and a separate configuration team to integrate and maintain these technologies. Collaboration with security teams to develop data privacy strategies is also recommended by P13. P18 (Machine Learning Engineer) highlighted the lack of recognition of the importance of clean large data sets despite their acknowledged role in improving model performance. This issue **stems from** the underestimation of the value of effective data collection and labeling. Lastly, P16 (Machine Learning Engineer) highlighted the resistance of managers to acknowledge suggested practices in the ML domain, which **stems from** the upper management's lack of technical knowledge in ML.

TABLE 4. Sub-category debt definitions for people category.

Debt Type	TD Definition
Technology and Knowledge Adaptation (9 cases)	***Resistance to new technology:*** Resistance to adapting new technologies which are essential for an ML project, but relying on obsolete systems ***Lagging behind developments:*** Inability to catch up with recent developments in the ML domain ***Rapid adoption of technologies:*** Rapidly adopting new ML project and technologies without adequate resources ***Management resistance to developments:*** Managers' resistance to acknowledging suggested ML practices ***Neglecting importance of an evolving standard:*** Insufficient recognition of the importance of clean, large datasets leading to better model performance and more reliable outcomes
Customer Expectation Alignment (2 cases)	***Expectation misalignment:*** Unrealistic expectations of stakeholders (customers) on outputs of ML projects
Culture (3 cases)	***Reluctance on raising issues:*** Reluctance of team members to raise concerns regarding seniors' work or discussing issues with upper management ***Internal resistance to ML team solutions:*** Solutions produced not being accepted within the organization
Knowledge Alignment (2 cases)	***Stakeholders disagreement:*** Stakeholders (project team and customers) do not agree on model inference results ***Project team disagreement:*** Challenges in choosing the best solution among conflicting of ML recommendations team members at the same role
Skill Integration (1 case)	***Challenges in integrating new team members:*** Difficulty for individuals with diverse industry experiences to adapt to and integrate into the team
Motivation (1 case)	***Reluctance for essential tasks:*** Unwillingness to perform critical ML tasks, such as data preprocessing, often accompanied by an eagerness to skip directly to modeling
Team Collaboration (4 cases)	***Failure to leverage expertise:*** Difficulty in determining who to collaborate with, or ML teams not leveraging the full range of expertise and perspectives available within the company ***Inadequate coordination:*** Not working together to track progress, leading to duplication of similar tasks ***Absence of senior guidance:*** Reluctance of senior staff to lead inexperienced team members
Team Communication (6 cases)	***Inefficient communication:*** Inefficient communication among ML teams and other stakeholders in the company ***Distortion of information:*** Information generated at lower levels of the team tends to get lost or distorted as it moves up to higher-level managers ***Conceptual language gap:*** Communication gap arising from jargon specific to ML and business domain
Management and Team Alignment (6 cases)	***Expectation misalignment:*** Unrealistic expectations of stakeholders (upper management, other relevant teams in the company) on outputs of ML projects ***Conflict over schedule and scope:*** Conflict between ML teams and upper management due to estimation (such as time and project scope) uncertainty ***Adopting hype-driven ML projects:*** Upper managers' pressure to adopt immature and hype ML development projects and technologies ***Shifting company goals:*** Challenge of aligning teams with the frequent changes in direction of organization ***Impractical hardware demands:*** Upper management setting unrealistic goals concerning hardware solutions
Team Structure (2 cases)	***Lagged team synchronization:*** Difficulty in technical coordination in globally distributed teams ***Split attention***: Divided attention and focus issues of employees holding multiple roles across different projects

2) KNOWLEDGE ALIGNMENT DEBT

This debt refers to misalignment caused by discrepancies in knowledge and understanding between stakeholders of an ML project. One case involves disagreement between stakeholders, particularly between ML team members and customers, regarding the model inference results. As explained by P4, the primary **root cause** of the issue was the conflict of interest between members of the ML team and customers, who tend to be reactive rather than proactive in identifying and resolving problems. P4 suggested selecting an Explainable AI (XAI) model to provide transparency in model decisions as a **suggested solution**,thereby helping stakeholders understand and agree on specific label selections. Other **suggested solutions** recommended by P4 were to improve stakeholder engagement during the product maintenance stage and to improve communication channels between stakeholders and integrate them into an iterative-incremental ML development process.

Another case involved difficulties in choosing the best solution among conflicting recommendations from members of the ML team who hold similar roles. P6 stated that the **root cause** of this issue as the diverse backgrounds of ML team members.

> *"Having people from different backgrounds added diversity to our team but also made it harder to find common ground. Aligning our approaches and finding a balance took time and it was challenging."*– P6 (Data Scientist)

As a **suggested solution**, P1 and P6 recommended guiding new disagreeable team members towards adaptation and fostering a collaborative environment that values diverse perspectives while working towards a common goal.

3) MANAGEMENT AND TEAM ALIGNMENT DEBT

This debt often arises from conflicts and misalignments between upper management and the expectations of the project stakeholders, including ML team members, upper management, and other relevant teams, and the realities of ML projects, where P9 highlighted the **root causes** as the estimation uncertainties in ML projects, such as time and

TABLE 5. Sub-category debt definitions for management category.

Debt Type	TD Definition
Business Development (1 case)	***Resource Capability Perception Feasibility Concern:*** The loss of potential clients due to concerns about the company's ability to successfully complete the project, particularly regarding adequate staffing levels and resource availability
Measurement and Estimation (1 case)	***Metric-objective misalignment:*** Misalignment between the chosen performance metrics and the evolving objectives or needs of the business and its users
Project Monitoring (1 case)	***Absence of project tracking:*** Lack of a project monitoring and task management approach
Human Resource Management (4 cases)	***Inefficient expertise allocation:*** Non-specialized team members burdened with specialized tasks due to limited resources ***Inadequate human resource allocation:*** Insufficient allocation of experienced personnel, such as senior staff, to critical roles ***Overburdened teams:*** Teams with excessive workloads caused by insufficient staffing, often expected to take on multiple roles beyond their primary duties. This often includes challenges in managing tasks such as data labeling, user interface design, and front-end implementation in ML projects
Stakeholder Requirement Management (1 case)	***Continuous requirement adjustments:*** High potential of scope creep due to constant required changes in ML project data and model such as new use cases and emerging and better ML models

scope, and the focus of the business side on product and deadline.

P13 noted another issue in which the upper managers pressure the ML teams to adopt immature and hyped ML development projects and technologies. This was **rooted in** upper management's tendency to overlook existing technology constraints and insist on staying competitive in the market, with a FOMO-driven impulse to continually experiment with new technologies. The **band-aid solution** to this issue involved forcibly adapting the company to the technology constraints imposed by the upper management.

> *"We are making architectural changes or UI adjustments to handle the project. This process often turns every step we take into a potential source of technical debt, affecting not only my team, but also the UI, DevOps, and production teams."*– P13 (Senior Manager – Search Engineering)

P13 also reported the difficulty of aligning teams with frequent changes in organizational direction. The **root cause** of this issue is upper management's eagerness to frequently change company goals due to swift improvements in the ML domain.

> *"We're expected to achieve our goals continuously in each quarter, but there can be multiple changes to the strategy, sometimes even twice in the same month. This requires us to redirect the entire team and shift focus from one task to another."*– P13 (Senior Manager – Search Engineering)

As noted by P10, P17, and P18, upper management sets unrealistic goals concerning solutions that are **rooted in** the limited understanding and knowledge of managers in the ML domain and their technical aspects such as hardware capabilities and constraints. A **suggested solution** would be to carry out a preliminary investigation to assess feasibility, avoiding the commitment of all resources when there is uncertainty regarding hardware integration.

> *"Upper management tends to have high expectations for ML projects. They are used to deterministic processes; therefore, when ML models produce unpredictable results, they find it difficult to understand."* – P18 (Machine Learning Engineer)

> *"If I'm asked to commit all resources, I prefer to allocate them initially to test the feasibility…Defining the lower and upper bounds of potential benefits before starting a project helps manage expectations."* – P17 (Optimization Team Lead/Senior Data Scientist)

The formation of a dedicated team to perform an in-depth analysis and the creation of a report to evaluate viability were also recommended by P17 as another **suggested solution**.

4) CUSTOMER EXPECTATION ALIGNMENT DEBT

This debt arises from the misalignment between customer expectations and the actual capabilities of ML projects. As noted by P9 and P14, the **root cause** of this is customers' inflated expectations of what ML and DS can deliver, imagining that these technologies can solve problems effortlessly.

> *"There is a perception that ML can magically solve all our problems, but that is not always the case. Customers are often very eager to implement ML solutions quickly, while the Data Science Team tends to be more cautious, advocating*

thorough consideration and planning." – P9 (Co-founder/CTO)

P14 also highlighted another primary **root cause**: customers' insufficient knowledge of the complexities and limitations inherent in ML technologies, placing additional pressure on ML teams to meet expectations that are not technically feasible.

The **suggested solution** would be to adopt a risk-aware commitment approach by the ML teams and thoroughly assess the feasibility of ML solutions during the initial phases of a project. This involves clear communication and setting realistic expectations, ensuring that customers are fully aware of the potential limitations of ML projects.

5) CULTURE DEBT

This debt arises from deep-rooted cultural issues within organizations that hinder progress and innovation in ML projects. One key issue involved the reluctance of team members to raise concerns regarding senior developers' work. The root cause of this was the hesitation of junior team members to challenge or question the work of seniors, mistakenly assuming it to be flawless, as noted by P4.

> *"There's this misconception among juniors that seniors' code is always optimal. Seniors can sometimes write hardcoded solutions because they are less open to new approaches, relying on what has worked for them in the past."*– P4 (Anomaly Detection Team Lead)

As a **band-aid solution**, P4 highlighted that instead of creating a proper code documentation, explanatory variable naming was used as a workaround. This solution was assumed to be sufficient to understand the code as it did not address the root cause. In this case, establishing a clear code review process that actively involves junior members, was **suggested as a solution**, as stated by P4.

> *"We encourage juniors to make corrections on the code they believe are necessary and then review their changes together, testing and validating the improvements."* – P4 (Anomaly Detection Team Lead)

Another challenge was the general unwillingness to discuss project issues with upper management. By P6, this reluctance was **rooted in** the frequent turnover of employees, and thus the management's tendency to invest in tools over human resources, which limits the growth opportunities of the ML team.

Finally, P10 highlighted how the solutions produced by the ML team were not readily accepted within the organization. The **root cause** of this was the absence of an adhocratic culture that emphasizes flexibility, innovation, and adaptability, thus altering familiar and comfortable habits, especially for a company with over 10,000 people (C8).

6) MOTIVATION DEBT

This category is characterized by a lack of enthusiasm and willingness among team members to engage in essential but less glamorous tasks, such as data preprocessing tasks. P3 explained the **root causes** of this reluctance as a desire to explore more exciting aspects of Data Science and ML, such as model building and a lack of real-world data challenges in the academic curriculum, leading to a mismatch between theory and practice.

> *"Recent graduates tend to bring an academic mindset in which they are eager to try new algorithms and experience the exciting and glamorous aspects of Data Science. They do not want to work with or thoroughly examine the data, but the solution to most problems lies in it."* – P3 (Data Scientist)

P3 **suggested** assigning task ownership to team members to foster a sense of responsibility and engagement as a **solution**. Encouraging seminar attendance was also **suggested** to help shift the mindsets of team members.

> *"What I've noticed is that people only start to understand when they take ownership of the problem and face it themselves."* – P3 (Data Scientist)

7) SKILL INTEGRATION DEBT

This debt arises from the challenges of merging team members with diverse industry experience into a cohesive team dynamic and adapting to the established workflows and practices of the team. This challenge is primarily **rooted in** persistent learned behaviors and difficulty in breaking away from their established patterns, as noted by P1.

> *"When we bring new team members on board, we try to be open-minded. If they have better approaches based on their previous experience, we are always willing to adopt them, but we also expect flexibility from them if our methods prove more effective."* – P1 (VP of Data)

To improve team integration, P1 **suggested** encouraging new team members to adapt to work methods and team culture as a **solution**.

8) TEAM COLLABORATION DEBT

This category is characterized by challenges associated with poor collaboration within and between teams. One case involved inadequate coordination between teams, leading to unnecessary duplication of similar tasks. P8 described the **root causes** as a lack of an incremental and iterative project life cycle, along with the failure of the project team to foster a culture of learning, sharing, and collaboration.

Another issue reported by P8 was the reluctance of senior staff to guide and mentor inexperienced team members. This reluctance is **rooted in** the organizational culture, which does not support or encourage mentor-mentee relationships.

P7 further highlighted the difficulties in identifying with whom to collaborate in large organizational structures. The **root cause** of this is that the complexity and size of the organization make it difficult for team members to identify relevant collaborators. In such cases, P7 **suggested** that mid-level managers navigate team members to relevant parties to facilitate better collaboration within the company.

Similarly, another case resulted in missed opportunities to leverage the expertise and perspectives available within the company. According to P10, this was **rooted in** the fact that ML teams often operated in isolation, developing complete solutions from start to finish, as stated below.

> *"Our ML team was so isolated that we had no idea what was happening in the company; we were living in a completely separate world. The negative impact of this is that we missed out on potential collaborations."* – P10 (Senior Computer Vision Engineer)

9) TEAM COMMUNICATION DEBT

This debt often arises from inefficiencies and misunderstandings in communication between project teams within a company, as articulated by P10, P11, P12, and P18. According to P11, the misunderstandings between the project team members were **rooted in** the diverse backgrounds and experiences of ML team members. P18 also stated that the **root cause** was the different interpretations and uses of key concepts and terminology among employees with diverse backgrounds.

> *"We face challenges with terminology. For example, when we mention 'network,' it's sometimes misunderstood as referring to physical wired networks, which leads to confusion."* – P18 (Machine Learning Engineer) implying neural networks but not physical networks in the statement.

Furthermore, the lack of senior stakeholders' knowledge in the ML domain contributes to NTD as another **root cause**, as these stakeholders may be technically unaware or misinterpret key concepts. As a **band-aid solution**, ML teams often took tasks outside their expertise to speed up project progress, as discussed by P11 and P12. A **suggested solution** would be to recognize the need for individuals from diverse backgrounds and roles to adjust to AI project workflows according to P11. P12 and P2 also recommended communicating well-prepared and clearly defined requirements and possible solutions to teams with insufficient ML knowledge. P12 also emphasized the importance of implementing temporary solutions by the ML team when the required team's involvement was not feasible and delegate tasks to the appropriate teams instead of owning the task.

Another significant issue reported by P10 is the tendency for the information generated at the lower levels of the team to be lost or distorted as it moves up the management hierarchy. This is **rooted in** the lack of transparency in communication between different layers of management.

> *"…Information had to pass through multiple layers—I'd explain something to my manager, who would pass it on, and then it would be relayed further. By the time the information reached higher levels, it often became lost or diluted."* – P10 (Senior Computer Vision Engineer)

As a **band-aid solution**, P10 noted that efforts to establish direct communication with the upper management as a workaround to avoid hierarchical information loss were ineffective.

P17 noted a communication gap that arises from the use of jargon specific to ML and business domains, which decreases stakeholder engagement. The **root cause** of this issue was the ML team's use of jargon to showcase their work and general reliance on technical terms.

> *"I guess that's human nature: people use jargon or buzzwords to make their work seem more important or specialized. However, this complicates the understanding for others."* – P17 (Optimization Team Lead / Senior Data Scientist)

To address this issue, P17 **suggested a solution** to encourage team members to simplify their explanations of technical subjects.

> *"I often encourage people to explain things as if they were talking to their grandparents, simplifying the explanation."*– P17 (Optimization Team Lead / Senior Data Scientist)

Additionally, leveraging expertise through collaborative documentation review can help team members better understand technical and domain-related terms.

10) TEAM STRUCTURE DEBT

This subdebt category is characterized by coordination challenges arising from organizational structure issues. In one case, the ML team had difficulties in technical coordination within globally distributed teams, which is **rooted in** working in different time zones and language barriers that slow decision-making and collaboration.

> *"When we send a question through company channels, if we are lucky, the person on the other end understands it correctly and provides an answer. However, if they do not understand, they respond to their own questions, and by the time we receive and reply, one day might already have passed. This back and forth can significantly slow things down, creating a time-based challenge for us."* – P13 (Senior Manager – Search Engineering)

Another case involved the divided attention and diluted focus of employees who held multiple roles in different projects. P13 discussed the **root cause** as decisions during portfolio-level strategic planning.

> *"Each team has its own agenda, and unfortunately, we have to work within a matrix structure that I never envisioned. In this setup, one person might act as a developer at one moment and then switch*

to a DevOps role, MLOps role, or even a tester role as needed. In our structure, a person might be working on project A, then shift to B, C, or D, and when things calm down in one area, they return to a previous project. This constant switching causes individuals to struggle to maintain their focus. " – P13 (Senior Manager – Search Engineering)

B. MANAGEMENT DEBT SUB-CATEGORIES AND RELATED CASES

In this section, we present the five debt categories identified under the Management Debt category: Measurement and Estimation Debt, Project Monitoring Debt, Human Resource Management Debt, Stakeholder Requirement Management Debt, and Business Development Debt. Similar to People Debt, each category includes cases with their root causes, along with descriptions of band-aid solutions and suggested solutions, if stated by the participants.

1) MEASUREMENT AND ESTIMATION DEBT

This debt is characterized by inaccuracies in project estimation and misalignment of performance metrics with business objectives. The **root causes** of this misalignment include metric blindness, where the project team takes the significance of metrics for granted and overlooks their limitations. An example of this is the overreliance solely on a "North Star" metric, which fails to capture the nuanced performance aspects required for an accurate evaluation of the ML project. The statement of P13 on the subject is as follows:

"I think there's a 'metric blindness' that develops when people become too used to certain industry-standard metrics. These metrics appear in a typical search or have been used for a long time in the sector. Over time, even if these metrics are complex in their own right, they start to appear trivial. This makes it difficult for people to accept new or different metrics." – P13 (Senior Manager – Search Engineering)

This case is also **rooted in** the reluctance of managers and project teams to adopt new metrics for performance measurement.

"The biggest challenge here was getting the management and product teams to understand the importance of measuring the right metrics. They often focused on metrics that did not align with our technical goals, rather than considering the relevance of metrics, such as classification accuracy, F measure, recall, or precision. This disconnect between technical metrics and business-focused metrics is a recurring difficulty for us." – P13 (Senior Manager – Search Engineering)

As a **suggested solution**, the ML team mathematically demonstrated that the company's North Star engagement metric did not adequately reflect the performance of the search algorithm. Developing a domain-specific metric, such as a 'search success ratio', could provide a more accurate and meaningful measure of algorithm effectiveness for such a project in the Educational Technology domain.

2) PROJECT MONITORING DEBT

This debt results from insufficient monitoring and management of project progress and tasks. P15 highlighted that ML teams do not have a structured project monitoring or task management approach. To address this, Kanban-like boards for tracking resources and tasks have been **suggested as a solution** for the project team.

"At first, we were managing everything in spreadsheets, but having a Kanban board would have made things much clearer and could have significantly speeded up our processes because it allows us to see exactly what everyone is doing and where all resources are at any given time." – P15 (Computational Linguist)

3) HUMAN RESOURCE MANAGEMENT DEBT

The issues subject to this debt are particularly related to the allocation and utilization of human resources. P13 described how the ML team struggled to efficiently manage the design and front-end tasks within the ML project at the current staff level. In addition to the lack of proper human resource allocation, another **root cause** of this issue was the frequent changes in the responsibilities and directives of the ML team, driven by the changes in company goals set by upper management.

As a **band-aid solution**, ML teams occasionally take initiative and perform front-end tasks outside the scope of their job descriptions or expertise. P13 details this issue as follows:

"Some of the team members have certain skills or 'superpowers' that we ultimately rely on, even if it's not technically their job. This is really a resource allocation problem and in our case, when we do not have enough designers or UI specialists, it can lead to bottlenecks." – P13 (Senior Manager – Search Engineering)

Another issue on overburdened teams noted by P14 is that the annotation team struggles to handle the increased workload of data labeling. This issue is **rooted in** the project expansion. Rapid growth made it difficult for the team to keep pace with demands.

"In terms of scalability, we initially managed with our in-house annotation team and prepared the data accordingly. However, as the project has grown significantly, it is no longer sufficient. We now need to label data from hundreds of sources and it is not feasible to quickly hire enough people to keep up with this demand." – P14 (Senior Deep Learning Engineer)

As a result, P14 **suggested a solution** to maintain the in-house labeling team while outsourcing the annotation process when necessary. This allowed the team to rapidly label the data and maintain a high level of efficiency.

Additionally, P3 reported the burden placed on non-specialized team members to handle specialized tasks due to limited resources. The **root causes** of this problem were a lack of recognition of the diverse skill sets required in ML projects and ambiguity in responsibility assignments.

> *"It is unusual for me to see a single person or team take on an entire project from start to finish. I think there could be more technical support, particularly on the Ops side. For example, we could have a developer from our team work alongside Ops to design a more robust system."* – P3 (Data Scientist)

Finally, P8 reported that the lack of senior staff in the ML team hindered the effective execution and oversight of ML projects.

4) STAKEHOLDER REQUIREMENT MANAGEMENT DEBT

This debt arises from managing and clarifying stakeholder requirements. P14 highlighted a high potential for scope creep due to constant changes required in data and models, which is **rooted in** the lack of clarity in project requirements, as the ML domain often demands continuous improvement.

> *"The problem is that the beginning and end of research are not clearly defined in ML. For example, if you are tasked with developing a detection model, you could theoretically continue to improve it indefinitely because there is no clear stopping point. Unlike software, where you can say that the code works, it has passed all tests and is bug-free. In ML, it is not always possible to determine when a satisfactory endpoint is reached. The project does not have a clear 'definition of done'."* – P14 (Senior Deep Learning Engineer)

Another **root cause** of this issue is that the process is frequently reset due to new use cases, new data, new methods, or emerging and better models, further complicating the management of stakeholder requirements. To address the high potential for scope creep, P14 **suggested** ensuring a good balance between the changes requested in the product requirements and the project constraints, including time and budget, as a **solution**. This approach may help manage stakeholder expectations and maintain the focus of the project.

5) BUSINESS DEVELOPMENT DEBT

This debt arises when strategic decisions and resource management inadequacies hinder a company's growth and client acquisition efforts. A specific case involved losing potential clients, which is **rooted in** their hesitation about the company's ability to successfully complete an ML project and the company's inability to adequately address customer concerns regarding its resource capacity, as can be seen in the statement of P18 below.

> *"Potential clients gave us feedback, saying that we needed more staff to get the job done effectively, and without increasing our team size, they would not be able to collaborate with us."* – P18 (Machine Learning Engineer)

These findings indicate that companies must prioritize proactive resource planning and transparent communication with potential clients to build confidence and secure new growth opportunities.

VI. DISCUSSION

This study offers novel insights into NTD in ML-based software development by introducing 10 People Debt and 5 Management Debt categories. It identifies the root causes, band-aid fixes, and suggested solutions, contributing new perspectives while supporting and extending prior research.

Team Structure Debt related issues were studied by John et al. [31]. They emphasized the importance of domain experts in addressing challenges in MLOps, such as those related to business cases, data, modeling, and deployment. Our study confirms this finding, revealing that employees holding multiple roles across projects within a matrix structure lead to diluted focus and inefficiency. This finding underlines the necessity of specialized roles, such as MLOps engineers or testers, in contrast to hub-like ML team structures. Granlund et al. [29] also highlighted the MLOps challenges in multi-organization setups, particularly the lack of coordination and standardization. Similarly, our study identified **Team Structure Debt** in globally distributed teams, where time zone differences and language barriers slowed decision-making and collaboration.

On the other hand, Munappy et al. [30] noted a lack of qualified professionals capable of both data analytics and DevOps as a major challenge, along with reluctance among data scientists to adopt new tools and technologies. Our study revealed similar results. In one of our cases, our participants mentioned the resistance to adopting new ML technologies, particularly from technical teams outside of ML teams, such as DevOps teams, which we classified as **Technology and Knowledge Adaptation Debt**. However, our findings also show the importance of fostering collaboration with technical teams to address challenges. The interview participants suggested that collaboration with DevOps teams is a suggested solution instead of relying solely on the shortcuts of the ML team, which can introduce security vulnerabilities. Our study also revealed challenges unique to ML-based projects, such as managerial resistance to ML practices and unrealistic hardware goals driven by limited ML knowledge. These findings highlight the need for domain-specific managerial expertise, which is not explicitly covered in the literature [17].

Both John et al. [31] and Bécue et al. [32] emphasized the scarcity and high cost of skilled professionals, such as data

scientists and subject matter experts, presenting these issues as challenges to acquiring and retaining talent. However, our findings diverge by focusing on the operational impact of insufficient staffing. Specifically, **Human Resource Management Debt** in our study was characterized by non-specialized team members being burdened with specialized tasks, lack of senior staff, and challenges in managing annotation workloads. In addition, inefficiencies in tasks, such as user interface design and front-end implementation, were attributed to inadequate staffing. This contrast suggests that the consequences of resource shortages in ML projects extend beyond financial considerations, affecting project execution and team efficiency.

Although Saeeda et al. [17] and Caballero-Espinosa et al. [24] discussed these issues in a broader context of software development, our findings emphasize how such challenges manifest themselves specifically in ML projects. Process debt, as identified in the study by Saeeda et al. [17], refers to challenges within software development processes that can delay or hinder specific development activities. The researchers highlight that a primary contributor to process debt is the reliance on shortcuts and quick fixes during the development process. Similarly, our study observed numerous instances of band-aid solutions within ML projects, such as ML teams bypassing the approval of DevOps teams for microservices, leading to security vulnerabilities. This aligns with the DevOps clash described by Caballero-Espinosa et al. [24], where tensions between the development and operations teams slow down the processes and lead to ineffective operations.

The influence of external dependencies, such as technology, tools and trends, was identified by Saeeda et al. [17] as another significant cause of process debt, noting how organizations often adopt processes that are not aligned with their needs due to external pressures such as global competition or volatile markets. Our study also highlights analogous challenges that are specific to ML projects. For example, as a **Technology and Knowledge Adaptation Debt**, Search Engineering Senior Manager emphasized the rapid adoption of new ML technologies without resource consideration, driven by the upper management's FOMO and competitive pressures. Additionally, in **Management and Team Alignment Debt**, upper managers pressured ML teams to adopt immature and hyped ML technologies, often ignoring existing constraints. Therefore, organizations forcibly adapted their structures to fit the imposed technology constraints, further compounding TD. Early adoption of tools has also been shown to reduce resource allocation issues and the long-term impact of TD [45].

Saeeda et al. [17] also noted that poor teamwork and coordination often impede workflow and prolong task completion, highlighting the impact of poor collaboration on project efficiency. Similarly, our study identified **Team Collaboration Debt**, where a Data Scientist we interviewed described cases in which inadequate coordination between teams led to unnecessary duplication of similar tasks. The root causes include the lack of an incremental and iterative project life cycle and the failure of the project team to foster a culture of learning, sharing, and collaboration.

Caballero-Espinosa et al. [24] defined social debt as a cumulative and increasing cost resulting from invisible and adverse effects within a development community, emphasizing how sociotechnical decisions and suboptimal social structures contribute to this debt. They further highlighted the relationship between social debt and TD, suggesting that technical issues such as flawed documentation or software components often arise from incorrect sociotechnical decisions. Similarly, a sociotechnical debt in our study was reflected as **Culture Debt**, exemplified by a case where junior team members were reluctant to raise concerns about the work of senior developers, due to the assumption that the work of senior developers was flawless. This behavior is closely aligned with the concept of cognitive distance described by Caballero-Espinosa et al. [24], where perceived hierarchical gaps hinder effective collaboration and exacerbate technical and social challenges within teams.

Caballero-Espinosa et al. [24] identified cognitive distance as a key factor in social debt, which arises from gaps between team members at the physical, technical, social and cultural levels, often leading to conflicts and inefficiencies. Saeeda et al. [17] also emphasized the importance of respecting diverse opinions and implementing conflict resolution strategies to mitigate social debt. In line with these insights, our study highlights **Skill Integration Debt** in ML projects driven by challenges in integrating team members with diverse industry experiences. Persisting in learned behaviors and resistance to established workflows hinder effective collaboration. As proposed by the VP of Data we interviewed, fostering flexibility and mutual adaptation can address this issue. This approach reinforces the importance of effectively managing diversity to build adaptable teams for ML projects.

Fischer et al. [28] noted the theory-practice gap in ML, where theoretical assumptions, such as data being drawn independently and identically distributed, often fail in real-world applications such as natural language processing and computer vision. Our study also identified **Motivation Debt** in ML projects driven by a mismatch between academic theory and practical challenges. Team members often lack enthusiasm for essential but less glamorous tasks, such as data pre-processing. The root causes include a preference for more exciting aspects of ML, such as model building, and the lack of exposure to real-world data challenges in the academic curriculum.

Regarding suggested solutions to mitigate TD, Freire et al. [13] suggested effective project planning activities, including monitoring, as a solution to address process and requirement debt. Saeeda et al. [17] emphasized the importance of follow-up assessments in mitigating process debt. In line with these perspectives, our study identified **Project Monitoring Debt** as a recurring issue in ML projects, arising from the lack of structured monitoring and task management approaches. To address this, the implementation of practices

such as Kanban boards was proposed as a suggested solution, allowing better oversight and coordination of tasks.

Freire et al. [13] proposed appropriate task allocation as a key strategy for mitigating process and requirement debt. This aligns with our study's findings on **Human Resource Management Debt**, where overburdened teams struggled to meet project demands, such as the rapid expansion of data labeling tasks due to project growth. Initially managed by an in-house team, the increased workload made it unsustainable to scale sufficiently quickly. As a solution, it was suggested that an in-house team should be maintained while outsourcing the annotation process when necessary.

Freire et al. [13] emphasized the importance of controlling and negotiating software requirements as a means of mitigating requirement debt. They stated practices, such as well-defined requirements, effective stakeholder communication, clear scope statements, requirement change tracking, and ensuring customer commitment. Our study also identified **Stakeholder Requirement Management Debt** in ML projects, driven by the dynamic nature of the domain. As noted by a Senior Deep Learning Engineer, the lack of clarity in project requirements and the frequent resetting of processes due to new use cases, data, or models created a high potential for scope creep. To address this, balancing requested changes with project constraints, such as time and budget, has been proposed. This aligns with the emphasis of the literature on structured requirements management, while highlighting the additional complexity introduced by ML-specific factors, such as the need for continuous improvement in models and data.

Fischer et al. [28] mentioned that the interpretability challenge is a critical aspect of trusted AI, highlighting the need for explainability to improve the understanding and acceptance of complex ML models. They noted that as model complexity increases, explainability becomes crucial, especially when addressing ethical, legal, and domain-specific questions. Our study identified **Knowledge Alignment Debt** in ML projects stemming from misalignment between stakeholders caused by differing interpretations of model inference results. As stated by a team leader on the Anomaly Detection Team, a key solution to this issue is the adoption of XAI models to provide transparency in model decisions, allowing stakeholders to better understand and align specific outputs, such as label selections.

Academic experts observed that certain suggested solutions proposed by the participants could inadvertently lead to additional technical debt. The following are the analyses of specific cases where the suggested solutions might inadvertently contribute to ongoing TD according to the academic experts in the study. The participants suggested that when the DevOps team resisted new architectures and technologies, the ML team should independently implement temporary solutions. However, academic experts have pointed out that this approach can lead to recursive TD. By bypassing the necessary team involvement, the ML team might inadvertently create a situation in which temporary solutions become permanent fixes, leading to a fragmented and unsustainable system. The suggested solution should involve fostering collaboration and expertise within the DevOps team, rather than avoiding it.

The participants proposed implementing temporary solutions in which the ML team handled tasks that were typically outside of their scope due to the lack of knowledge of senior stakeholders in the ML domain. However, academic experts have identified this approach as a recursive TD solution. Continuously delegating tasks to the ML team that should be handled by other specialized teams can lead to overburden and inefficiencies. The suggested solution should involve joint training sessions for the software and ML teams to foster a better understanding of the workflows, challenges, and technical requirements of each other, and to hire cross-disciplinary professionals to bridge the knowledge gap and promote better collaboration.

In addition, companies can enhance their product development capabilities by adopting frameworks or standards such as CMMI [46], [47], which has been successfully applied in the software engineering domain for many years to ensure that organizations have the necessary capabilities to develop products effectively. To achieve this in the AI domain, it is crucial to establish clear standards and audit mechanisms specifically tailored to AI projects, such as those designed for AI auditing [48].

VII. THREATS TO VALIDITY

Potential validity threats were identified and addressed to ensure the reliability and generalizability of our findings. The following validity threats discuss the construct, internal, and external validity considerations relevant to this research study [49].

To maintain construct validity, we deliberately avoided using the term 'technical debt' during interviews, instead asking participants about the 'issues' they faced in ML projects. This minimized bias and allowed for responses grounded in real experience, which was particularly important for participants unfamiliar with the term. To ensure relevance, we focused on projects deployed in the last two years. We covered the full ML project life cycle outlined in TDSP [5] using open-ended questions to explore TD and its root causes. This approach enables the identification of categories across phases, including challenges involving external stakeholders.

Internal validity was supported through a structured research process, particularly during iterative artifact development and expert review. Multiple evaluation rounds allowed us to refine the categories and structure based on expert feedback, ensuring alignment with the collected data. The interview questions were carefully designed, reviewed by multiple researchers, and tested through a pilot interview to minimize bias and enhance clarity. Data saturation was determined when sufficient insights were obtained to meet research objectives and ensure comprehensive findings.

External validity was strengthened by including participants from various sectors, ML domains, company sizes, and geographic regions, helping avoid bias from personal networks. We also ensured a wide range of roles from executives to mid-level managers to capture varied perspectives on TD in ML projects. While further validation is needed across all domains, efforts have been made to create a descriptive and generalizable categorization. Additionally, five participants came from the same company, raising the possibility of repeated organizational insights. To address this, we ensured role diversity ranging from the VP of Data to Data Scientist to capture varied hierarchical perspectives.

VIII. CONCLUSION

This study categorizes TD related to people and management aspects in ML-based projects by integrating insights from various stages of the entire ML project life cycle. An in-depth root cause analysis was conducted to uncover how NTD emerges and interacts with its underlying causes across different life cycle stages. The analysis also identifies suboptimal decisions that lead to recurring forms of TD, as well as ideal practices that can help prevent or mitigate TD. Using DSR methodology, we conducted interviews with 18 industry professionals and applied thematic analysis to identify 15 distinct categories of TD, including issues such as resistance to change, misaligned expectations, and communication barriers. Our findings underscore the impact of several causes of NTD such as inadequate resource allocation, shifting priorities, and poor alignment between business goals and project objectives. Our findings have several implications for research and practice. For practitioners, categorization provides actionable insights into key NTD drivers. For researchers, this study establishes a foundation for future work to validate and extend the proposed NTD categorization across diverse industries and geographies, enhancing its generalizability and applicability. Furthermore, it highlights the potential for developing automated tools and frameworks to integrate TD management into MLOps practices, streamlining the identification and mitigation of TD and NTD. Evaluating the TD structure in real-world scenarios will further demonstrate its practical impact, particularly in improving project outcomes such as budget adherence, timeline compliance, and resource efficiency.

APPENDIX

INTERVIEW QUESTIONS

Introduction: The researcher introduces and provides information on the scope of the study.

Section 1: Please briefly answer these questions.

1) Please state your full name.
2) What is your major and the highest level of education you have completed?
3) Please specify your position and name of the department in which you work in your current organization.
4) Do you have roles other than the position you mentioned? What is your technical role?
5) Can you briefly describe the responsibilities of your position?
6) Please state your work experience in years.
7) Please state your work experience in the Data Science field in years.
8) How many Data Science or Machine Learning projects are you currently working on?
9) How many Data Science or Machine Learning projects have you completed throughout your career?
10) Are you familiar with the concept of technical debt?

Section 2: Answering these questions is optional.

1) Please specify the sector in which your company operates.
2) How many people work at your company? You can choose from the following ranges: (1-50 people, 51-200 people, 201-500 people, 501-1000 people, 1001-5000 people, and more than 5000 people).
3) Please state the country where the company is headquartered.
4) Please list any certificates obtained in the field of Data Science or in other fields.

Section 3:

1) Please state the name of the project to discuss the challenges.
2) How many people have worked on or are currently working on the project? Please specify their roles here.
3) How long did the project last? Please specify the start and end dates of the project.
4) Are the scope, budget, and timeline of your contract subject to defined and strict rules?
5) If a budget is planned and to your knowledge, what is the budget of the project?
6) Please specify the domain, purpose, and scope of the project.
7) In which project processes were you actively involved? (e.g., Deployment)
8) Did you follow a project cycle or a process management approach to solve the issues encountered at this stage?

Section 4

Data Preparation

1) Please describe the issues encountered during the Data Preparation phase of the project development process you mentioned and the possible root causes of these issues. If yes:
2) Did you apply shortcuts or suboptimal solutions to the issues you have mentioned at this stage?
3) (If solutions were mentioned) Were there any other shortcuts or suboptimal solutions that you did not mention or skip?
4) What are the managerial and technical root causes for applying these solutions? (Such as time pressure.)
5) What practices, methods, or tools did you use?
6) Did you apply an ideal (better) solution? If not, do you plan to resolve it?

7) What methods and practices did you observe that you did well in this process and wanted to continue doing? If no:
8) What were the reasons for not experiencing these issues?
9) What methods and practices did you observe that you did well in this process and wanted to continue doing?

Model Development

1) Please describe the issues encountered during the Model Development phase of the project development process you have mentioned and the possible root causes of these issues. If yes:
2) Did you apply shortcuts or suboptimal solutions to the issues you mentioned at this stage?
3) (If solutions were mentioned) Were there any other shortcuts or suboptimal solutions that you did not mention or skip?
4) What are the managerial and technical root causes for applying these solutions? (Such as time pressure.)
5) What practices, methods, or tools did you use?
6) Did you apply an ideal (better) solution? If not, do you plan to resolve it?
7) What methods and practices did you observe that you did well in this process and wanted to continue doing? If no:
8) What were the reasons for not experiencing these issues?
9) What methods and practices did you observe that you did well in this process and wanted to continue doing?

Model Deployment

1) Please describe the issues encountered during the Model Deployment phase of the project development process you mentioned and the possible root causes of these issues. If yes:
2) Did you apply shortcuts or suboptimal solutions to the issues you have mentioned at this stage?
3) (If solutions were mentioned) Were there any other shortcuts or suboptimal solutions that you did not mention or skip?
4) What are the managerial and technical root causes for applying these solutions? (Such as time pressure.)
5) What practices, methods, or tools did you use?
6) Did you apply an ideal (better) solution? If not, do you plan to resolve it?
7) What methods and practices did you observe that you did well in this process and wanted to continue doing? If no:
8) What were the reasons for not experiencing these issues?
9) What methods and practices did you observe that you did well in this process and wanted to continue doing?

Software and Hardware Tool Management and Configuration Management

1) Please describe the issues you encountered during the Software and Hardware Tool Management and Configuration Management of the project development process you mentioned and the possible root causes of these issues. If yes:
2) Did you apply shortcuts or suboptimal solutions to the issues you mentioned at this stage?
3) (If solutions were mentioned) Were there any other shortcuts or suboptimal solutions that you did not mention or skip?
4) What are the managerial and technical root causes for applying these solutions? (Such as time pressure.)
5) What practices, methods, or tools did you use?
6) Did you apply an ideal (better) solution? If not, do you plan to resolve it?
7) What methods and practices did you observe that you did well in this process and wanted to continue doing? If no:
8) What were the reasons for not experiencing these issues?
9) What methods and practices did you observe that you did well in this process and wanted to continue doing?

People, Culture, and Strategy

1) Please describe the issues you encountered during the People, Culture, and Strategy of the project development process you mentioned and the possible root causes of these issues. If yes:
2) Did you apply shortcuts or suboptimal solutions to the issues you have mentioned at this stage?
3) (If solutions were mentioned) Were there any other shortcuts or suboptimal solutions that you did not mention or skip?
4) What are the managerial and technical root causes for applying these solutions? (Such as time pressure.)
5) What practices, methods, or tools did you use?
6) Did you apply an ideal (better) solution? If not, do you plan to resolve it?
7) What methods and practices did you observe that you did well in this process and wanted to continue doing? If no:
8) What were the reasons for not experiencing these issues?
9) What methods and practices did you observe that you did well in this process and wanted to continue doing?

Strategic Alignment

1) Please describe the issues encountered during the Strategic Alignment of the project development process you mentioned and the possible root causes of these issues. If yes:
2) Did you apply shortcuts or suboptimal solutions to the issues you have mentioned at this stage?
3) (If solutions were mentioned) Were there any other shortcuts or suboptimal solutions that you did not mention or skip?
4) What are the managerial and technical root causes for applying these solutions? (Such as time pressure.)
5) What practices, methods, or tools did you use?

6) Did you apply an ideal (better) solution? If not, do you have a plan to resolve it?
7) What methods and practices did you observe that you did well in this process and wanted to continue doing? If no:
8) What were the reasons for not experiencing these issues?
9) What methods and practices did you observe that you did well in this process and wanted to continue doing?

Organizational Collaboration

1) Please describe the issues you encountered during the Organizational Collaboration of the project development process you mentioned and the possible root causes of these issues. If yes:
2) Did you apply shortcuts or suboptimal solutions to the issues you have mentioned at this stage?
3) (If solutions were mentioned) Were there any other shortcuts or suboptimal solutions that you did not mention or skip?
4) What are the managerial and technical root causes for applying these solutions? (Such as time pressure.)
5) What practices, methods, or tools did you use?
6) Did you apply an ideal (better) solution? If not, do you plan to resolve it?
7) What methods and practices did you observe that you did well in this process and wanted to continue doing? If no:
8) What were the reasons for not experiencing these issues?
9) What methods and practices did you observe that you did well in this process and wanted to continue doing?

Others: Are there any other problems you would like to mention that do not fit the mentioned phases? What practices, methods, or tools did you use to solve these problems?

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- We do not generate any datasets. Our work uses interview data, which can be found in [50].

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